

Leader Harmonious Drive Systems Co. (688017.SS)

Rising domestic humanoid demand with capacity expansion on the way; Testing valuation sensitivity on what's priced in

688017.SS 12m Price Target: **Rmb138.00** Price: **Rmb488.00** Downside: **71.7%**

We hosted a group conference call with LeaderDrive Board Secretary Mr. Gui on Jul 3, 2026. **Key takeaways:**

1) Vertical or dedicated applications (such as warehouse & logistics, packaging in supermarkets etc) could commercialize soon and drive 2026E total humanoid robot industry shipments to above 50k units (GSe base case at 51k);

2) Potential “EV-like” subsidies policy, according to the company, could catalyze a surge toward 200k units in 2027E, vs GSe base/bull/blue sky scenario of 76k/148k/220k if it materializes (or close to GSe blue sky scenario);

3) As LeaderDrive has a dominant market share of 80-90% (vs. 70% last year) among China humanoid robots that adopted harmonic reduction gears inside, the company commented that it's currently in tight capacity and is likely itself a key bottleneck of industry growth;

4) Pricing trend is stable this year, accordingly, vs. a usually declining trend in history. Leaderdrive did not plan to hike the price, in order to keep competition away and maintain its dominant positioning;

5) The company continues to reduce unit capex and ramp up capacity, aiming to realize an annualized harmonic gear capacity at 1.44mn units by end-2026 and 10mn units in the long term. Note that for a high DOF humanoid robot, more than 10 harmonic reduction gears might be needed per robot.

What's priced in? LeaderDrive share price has performed strongly at +154% ytd with the stock now trading at 411x/276x 2026E/2027E P/E against earnings growth of 68%/46% yoy, and GSe is already above Wind consensus by 9%/11%, which in our view suggests the stock is likely trading on long run prospects.

We test: 1) a range of 2030E shipment volume assuming the

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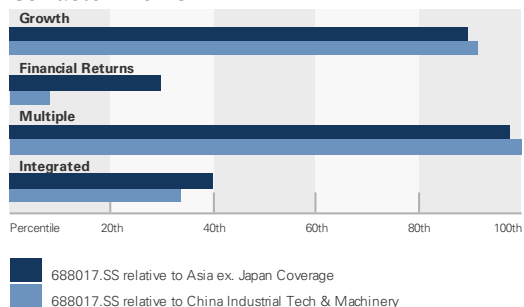
Key Data

Market cap: Rmb89.5bn / \$13.2bn
Enterprise value: Rmb89.0bn / \$13.1bn
3m ADTV: Rmb4.4bn / \$643.5mn
China
China Industrial Tech & Machinery
M&A Rank: 3
Leases incl. in net debt & EV?: No

GS Forecast

	12/25	12/26E	12/27E	12/28E
Revenue (Rmb mn)	570.7	896.4	1,196.5	1,536.1
EBITDA (Rmb mn)	143.8	224.5	331.5	455.4
EPS (Rmb)	0.68	1.14	1.67	2.32
P/E (X)	NM	NM	NM	NM
P/B (X)	7.6	24.2	22.9	21.2
Dividend yield (%)	0.1	0.1	0.1	0.1
N debt/EBITDA (ex lease,X)	(5.2)	(2.1)	(1.4)	(0.9)
CROCI (%)	7.9	9.7	11.8	14.0
FCF yield (%)	0.2	(0.3)	0.1	0.1
	3/26	6/26E	9/26E	12/26E
EPS (Rmb)	0.18	0.27	0.34	0.35

GS Factor Profile



Source: Company data, Goldman Sachs Research estimates.
See disclosures for details.

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Leader Harmonious Drive Systems Co. (688017.SS)

Rating since Jan 13, 2025

Ratios & Valuation

	12/25	12/26E	12/27E	12/28E
P/E (X)	NM	NM	NM	NM
P/B (X)	7.6	24.2	22.9	21.2
FCF yield (%)	0.2	(0.3)	0.1	0.1
EV/EBITDAR (X)	181.7	396.4	268.5	195.5
EV/EBITDA (excl. leases) (X)	181.7	396.4	268.5	195.5
CROCI (%)	7.9	9.7	11.8	14.0
ROE (%)	3.6	5.8	8.1	10.5
Net debt/equity (%)	(21.0)	(12.6)	(11.9)	(10.2)
Net debt/equity (excl. leases) (%)	(21.0)	(12.6)	(11.9)	(10.2)
Interest cover (X)	21.2	38,562.2	63,322.4	90,489.9
Days inventory outst, sales	169.7	151.8	165.0	168.9
Receivable days	153.7	129.0	121.2	115.6
Days payable outstanding	95.6	96.8	104.5	106.6
DuPont ROE (%)	3.5	5.7	7.8	10.1
Turnover (X)	0.1	0.2	0.3	0.3
Leverage (X)	1.1	1.1	1.2	1.2
Gross cash invested (ex cash) (Rmb)	2,596.0	3,151.3	3,529.0	4,073.5
Average capital employed (Rmb)	2,406.2	3,011.8	3,334.5	3,612.2
BVPS (Rmb)	19.33	20.14	21.33	22.97

Growth & Margins (%)

	12/25	12/26E	12/27E	12/28E
Total revenue growth	47.3	57.1	33.5	28.4
EBITDA growth	82.9	56.2	47.7	37.4
EPS growth	103.8	68.0	46.4	39.0
DPS growth	93.4	62.8	46.4	39.0
EBIT margin	14.7	15.3	18.9	21.0
EBITDA margin	25.2	25.0	27.7	29.6
Net income margin	21.8	23.3	25.6	27.7

Price Performance



Source: FactSet. Price as of 3 Jul 2026 close.

Income Statement (Rmb mn)

	12/25	12/26E	12/27E	12/28E
Total revenue	570.7	896.4	1,196.5	1,536.1
Cost of goods sold	(360.1)	(579.7)	(781.9)	(1,007.1)
SG&A	(39.7)	(31.4)	(15.6)	(14.6)
R&D	(53.9)	(98.6)	(119.6)	(138.2)
Other operating inc./exp.)	(33.0)	(49.3)	(53.8)	(53.8)
EBITDA	143.8	224.5	331.5	455.4
Depreciation & amortization	(59.7)	(87.1)	(105.9)	(133.1)
EBIT	84.1	137.4	225.6	322.3
Net interest inc./exp.)	(1.1)	9.7	6.1	6.1
Income/(loss) from associates	(0.8)	(0.2)	—	—
Pre-tax profit	144.3	215.7	319.9	449.3
Provision for taxes	(18.4)	(6.8)	(14.0)	(24.2)
Minority interest	(1.5)	—	—	—
Preferred dividends	—	—	—	—
Net inc. (pre-exceptionals)	124.4	208.9	305.9	425.1
Post-tax exceptionals	—	—	—	—
Net inc. (post-exceptionals)	124.4	208.9	305.9	425.1
EPS (basic, pre-exception) (Rmb)	0.68	1.14	1.67	2.32
EPS (diluted, pre-exception) (Rmb)	0.68	1.14	1.67	2.32
EPS (basic, post-exception) (Rmb)	0.68	1.14	1.67	2.32
EPS (diluted, post-exception) (Rmb)	0.68	1.14	1.67	2.32
DPS (Rmb)	0.21	0.34	0.50	0.70
Div. payout ratio (%)	31.0	30.0	30.0	30.0

Balance Sheet (Rmb mn)

	12/25	12/26E	12/27E	12/28E
Cash & cash equivalents	745.5	466.5	466.5	429.9
Accounts receivable	265.2	368.4	426.1	547.1
Inventory	285.0	460.6	621.2	800.2
Other current assets	1,296.0	1,296.0	1,296.0	1,296.0
Total current assets	2,591.7	2,591.5	2,809.8	3,073.2
Net PP&E	610.4	856.9	945.8	1,080.3
Net intangibles	56.8	49.6	42.5	35.3
Total investments	57.2	104.9	172.1	266.1
Other long-term assets	613.1	613.1	613.1	613.1
Total assets	3,929.1	4,216.1	4,583.3	5,068.0
Accounts payable	117.0	190.6	257.1	331.1
Short-term debt	0.1	0.1	0.1	0.1
Short-term lease liabilities	—	—	—	—
Other current liabilities	153.8	177.9	207.0	242.8
Total current liabilities	270.9	368.6	464.2	574.0
Long-term debt	0.0	0.0	0.0	0.0
Long-term lease liabilities	—	—	—	—
Other long-term liabilities	115.0	155.3	209.6	283.0
Total long-term liabilities	115.0	155.3	209.6	283.0
Total liabilities	385.9	523.9	673.8	857.0
Preferred shares	—	—	—	—
Total common equity	3,536.8	3,685.8	3,903.1	4,204.7
Minority interest	6.4	6.4	6.4	6.4
Total liabilities & equity	3,929.1	4,216.1	4,583.3	5,068.0
Net debt, adjusted	(745.4)	(466.4)	(466.4)	(429.8)

Cash Flow (Rmb mn)

	12/25	12/26E	12/27E	12/28E
Net income	124.4	208.9	305.9	425.1
D&A add-back	59.7	87.1	105.9	133.1
Minority interest add-back	1.5	—	—	—
Net (inc)/dec working capital	(3.5)	(205.2)	(151.9)	(225.9)
Other operating cash flow	(30.1)	(7.5)	(12.8)	(20.7)
Cash flow from operations	152.0	83.3	247.1	311.7
Capital expenditures	(99.7)	(326.5)	(187.7)	(260.4)
Acquisitions	—	—	—	—
Divestitures	—	—	—	—
Others	(662.5)	—	—	—
Cash flow from investing	(762.3)	(326.5)	(187.7)	(260.4)
Repayment of lease liabilities	—	—	—	—
Dividends paid (common & pref)	(22.4)	(38.5)	(62.7)	(91.8)
Inc/(dec) in debt	(148.1)	—	—	—
Other financing cash flows	1.5	2.7	3.3	3.9
Cash flow from financing	(168.9)	(35.8)	(59.4)	(87.8)
Total cash flow	(779.2)	(278.9)	(0.1)	(36.6)
Free cash flow	52.3	(243.2)	59.3	51.3

Source: Company data, Goldman Sachs Research estimates.

company continues to run at full UTR of its mid-to-long term capacity of 2~10 mn units (GSe base case at 2.5mn units by 2030E); and 2) an ASP range of Rmb700 (single harmonic reduction gear) to Rmb2,500 (integrated actuator module), **which gives a range of equity valuations that imply 67% downside ~ 147% upside to current share price** (Exhibit 1), suggesting the market is factoring in a bull case scenario.

Exhibit 1: Based on our sensitivity analysis, the implied up/downside to current share price ranges from -67% to 147%, assuming 2030E shipment volumes of 2~10 mn units (vs. GSe base case at 2.5mn units by 2030E); and a range of ASP from Rmb700 (single harmonic reduction gear) to Rmb2,500 (integrated actuator module) for humanoid-related products

Implied LeaderDrive equity valuation (Rmb bn)		ASP of humanoid-related products (i.e. harmonic reduction gears and integrated modules) (Rmb per unit, 2030E)						
Humanoid-related product shipment (mn units, 2030E)		700	1,000	1,300	1,600	1,900	2,200	2,500
	2	29	34	39	44	49	54	58
	4	41	50	60	70	80	89	99
	6	52	67	81	96	110	125	140
	8	63	83	102	122	141	161	180
	10	75	99	123	148	172	196	221

Implied LeaderDrive equity valuation (Rmb bn)		ASP of humanoid-related products (i.e. harmonic reduction gears and integrated modules) (Rmb per unit, 2030E)						
Humanoid-related product shipment (mn units, 2030E)		700	1,000	1,300	1,600	1,900	2,200	2,500
	2	-67%	-62%	-56%	-51%	-46%	-40%	-35%
	4	-55%	-44%	-33%	-22%	-11%	0%	11%
	6	-42%	-26%	-9%	7%	23%	40%	56%
	8	-29%	-7%	14%	36%	58%	80%	101%
	10	-17%	11%	38%	65%	92%	120%	147%

Base case
Bull case
Blue sky

Other key assumptions include NPM at 25% for humanoid related business; 45x 2030E exit multiple; CoE at 11.5%

Source: Goldman Sachs Global Investment Research

Detailed takeaways:

Surging humanoid robot demand in China with potential policy stimulus: Per management observations, humanoid robots are poised for deployment across specific dedicated applications in the near term, focusing primarily on three areas: **(1) material handling:** Enabling the flexible grasping, placement, loading/unloading of unstructured parts in industrial settings; **(2) logistics sorting:** addressing China's domestic market of 1.2 million sorting jobs; **(3) supermarket and retail operations:** covering tasks such as item packaging and shelf stocking. As a result, management believes **total humanoid robot shipment in China could exceed 50k units in 2026**. However, the company noted that highly precise operational applications are not yet mature, with the primary bottleneck lying in the insufficient reliability and lifespan of key components within dexterous hands (such as tendons). Consequently, commercial deployments at this stage remain reliant on conventional grippers. **Into 2027, company thinks the domestic industry shipment could potentially reach c.200k units, noting that the Chinese government is considering to launch a major policy in 2026** — potentially providing subsidies to the humanoid robot industry, similar to those for EV industry in the 2010s. For context, China launched the first batch of EV subsidy programs in 2009-10, offering up to Rmb60k/50k for the individual buyers of BEV/PHEV and up to Rmb500k subsidies for the buyers of EV buses in pilot cities, or up to 20%~40% of ASP depending on the car model ([link](#)).

Dominant domestic market share with full utilization and stable pricing:

LeaderDrive continues to run at a full utilization rate given robust demand. Management expects their market share in the domestic harmonic reduction gear market for humanoid robot applications has increased to 80%~90% (from ~70% in 2025) as many customers rely more on high-quality suppliers like LeaderDrive with strong aftersales and service abilities, as they transition from R&D to early commercialization. On ASP trend, pricing for harmonic reduction gears remains relatively stable in 2026 (vs. it has

been declining yoy in the past few years) given the tight supply situation, but in the longer term LeaderDrive still aims to bring down their pricing and GPM in order to discourage new entrants and maintain its dominant market position.

Aggressive capacity expansion to meet rising demand: Capacity has been ramping up rapidly in the past few months reaching a monthly capacity of c.70k units in May and c.80k units in June, and management targets to bring its monthly capacity of harmonic reduction gears to 120k units **by end-2026, implying 1.44mn units on an annualized basis vs. 600k-700k by end-2025**. While equipment and component supply is getting tighter, with the lead time of some high-end machine tools extended to 12 months, it is faster and smoother for LeaderDrive to increase capacity given their much larger scale and established supply chain relationships than the smaller peers. Unit capex has reduced materially in the past few years from ~Rmb1,000 per unit and is trending towards <Rmb500 per unit going forward. As such, the company will continue to develop more in-house manufacturing equipment, including CNC machine tools, milling machines, and grinding machines etc. **In the long term, the company envisions an annualized capacity of 10mn units**, but the ramp-up trajectory will depend on demand growth, domestic substitution on production equipment, manufacturing know-how improvements etc.

New JV with SKF (SKFb.ST, Covered by Daniela Costa): On Jul 2, SKF and LeaderDrive announced a plan to establish a JV in China ([link](#)) with 60%/40% ownership respectively, which will focus on high-precision transmission components for humanoid robots. The JV will start operations in end-2026, serving not only Chinese customers but also international markets such as Europe, Japan, and the US, leveraging SKF's global sales network. With this partnership, SKF will supply to LeaderDrive a series of products for humanoid applications, including crossed roller bearings, flexspline bearings, and deep groove ball bearings, and they will collaborate on the lightweight design and manufacturing techniques of harmonic reduction gears.

Not yet seeing concrete large-scale orders from US humanoid customers: In the North American market, the company continues to closely work with several key customers on incremental refinements on harmonic reduction gear designs regarding back-drivability, noise reduction, and life span etc. Multiple overseas customers are asking LeaderDrive to supply integrated rotary modules (with reducers, motors, force sensors, encoders etc.), which typically costs Rmb2,000~3,000 per module, and the company sees room for further reliability enhancement and cost reduction for module products. That said, no clear acceleration of demand has been observed, as the company is still waiting for concrete orders rather than relying on customers' volume forecasts. Separately, management also sees rising interest from European customers mainly related to logistics applications.

Progress with global big4 industrial robot brands: LeaderDrive continues to gain traction with the top 4 international industrial robot brands: LeaderDrive is a major supplier of harmonic reduction gears with on-par quality and more competitive pricing vs. global peers: with Yaskawa and ABB, the company has achieved small batch volume shipment, likely for their industrial robots sold in the Chinese market; for Fanuc, after testing for almost 1.5 years, LeaderDrive plans to set up a sales channel with Fanuc in Japan in 2026.

Other transmission components: For planetary roller screws, the company is seeing

limited adoption in the domestic humanoid robot market; instead, it is working with some equipment customers and targets <Rmb100mn sales in 2026. For planetary reducers, LeaderDrive will likely start volume delivery in 2H26 after going through testing with several key customers in 1H26.

The author would like to thank Zhou Li, Hao Chen, Zhihan Ye, and Junfang Zhang for their contribution to this report.

Investment thesis, valuation and risks

Investment thesis: On valuation, we are Neutral rated on LeaderDrive, the domestic leader in China's harmonic reduction gear market. We are positive on the company for its growing harmonic reduction gear TAM expansion, due to a growing set of applications both domestically and globally across industrial robots, collaborative robots (cobots), humanoid/service robots, CNC machine tools and other high-end equipment (solar/SPE/medical/aerospace). The company is in the process of penetrating into overseas customers, such as making volume shipments to 3 of the Big 4 international robot brands and expects more meaningful revenue contribution into the next few years.

Price Target, Risks and Methodology: Our 12-month target price for Leader Harmonious Drive Systems Co. is Rmb138.0, based on a 2030E P/E of 45x, discounted back to 2027E at an 11.5% CoE. Upside risks: (1) faster-than-expected volume production and/or technology advancement of humanoid robots; (2) faster-than-expected penetration into key robot customers (especially foreign brands). Downside risks: (1) worse-than-expected domestic industrial/collaborative robot demand; (2) more competition pressure from overseas and local brands.

Disclosure Appendix

Reg AC

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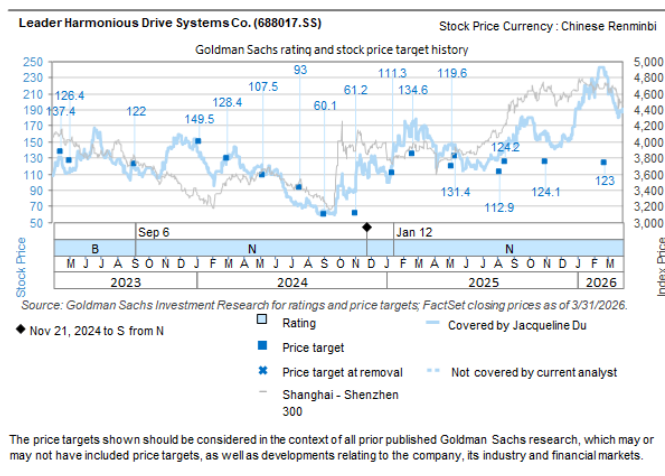
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Price target and rating history chart(s)



Target price history table(s)

Leader Harmonious Drive Systems Co. (688017.SS)

Date of report Target price (Rmb) Closing price (Rmb)

23-Apr-26	138.00	203.77
24-Feb-26	123.00	232.70
03-Nov-25	124.10	159.50
17-Aug-25	124.20	148.31
06-Aug-25	112.90	140.13
13-May-25	131.40	146.06
07-May-25	119.60	146.07
20-Feb-25	134.60	169.83
12-Jan-25	111.30	120.04
04-Nov-24	61.20	106.20
05-Sep-24	60.10	65.04
18-Jul-24	93.00	71.70
09-May-24	107.50	115.31
01-Mar-24	128.40	129.88
08-Jan-24	149.50	131.96
06-Sep-23	122.00	117.20

Price targets shown in table(s) are unadjusted for corporate actions.

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